DATASCI 406 FA 26, Quiz 1.5

Name and Uniqname:

Question 1 (4 points)

Using today's exercises, identify an example of each of these techniques we have covered in this module.

(a) Estimating a feature of a distribution (such as an expectation or probability) for which you do not have a closed form answer.

(b) Finding the operating characteristic of an estimator or testing procedure.

(c) Approximating the sampling distribution of a statistic that has no tractable form.

(d) Investigating a claim you have no formal proof of.

Solution

(a) The p-value.

(b) The power against `p_alt`.

(c) The 4,000 simulated distance statistics — the null distribution.

(d) That F becomes uniform as k grows, or that $F \sim U(0,1)$ gives digits at w_d.

Rubric. 1 pt each, generously. Any exercise fitting the description counts, and one exercise may serve more than one part — the null simulation is a fair answer to both (a) and (c).

Question 2 (1 point)

The paper's conclusion mentions tests built on the **second** digit rather than the first. To run today's test that way you would need the distribution of the second digit under the null.

Assume Benford's Law still holds — that is, $F \sim U(0,1)$. Explain how you would use a Monte Carlo technique to estimate the probability of second digits $d_2 = 0, \ldots, 9$. You may assume a function `second_digit()` exists already.

Solution

The generation exercise, reading a different digit. Draw $F \sim U(0,1)$, set $X = 10^F$, and tabulate the second digit. Code might look like this:

```
x <- 10^runif(n)
table(factor(second_digit(x), levels = 0:9)) / n
```

M can be dropped entirely. It moves the decimal point and changes no digit, which is why the leading digit depended only on F in the first place. And no new theory is needed: assuming F uniform already fixes every digit, not just the first.

Rubric. 1 pt for simulating $F \sim U(0,1)$ and reading the second digit. Keeping M is fine. No credit for "look up the second-digit law."